

On the Expressive Limits of Formal Truth Predicates

Alex Doe

Munich Center for Mathematical Philosophy

alex.doe@lmu.de

ABSTRACT

We re-evaluate the axiomatic theories of truth extending Peano Arithmetic. By restricting reflection principles to grounded syntactic formulae, we preserve classical logic while resisting Tarskian semantic hierarchies.

Keywords: Axiomatic Truth, Kripke-Feferman, Incompleteness, Model Theory

MSC (2020): 03C90, 03B42

1. Introduction

Formal theories of truth face classical diagonal limits, as originally formulated by Tarski (1936). A recurrent difficulty concerns whether typed semantic structures are mandatory to avoid paradox ([Halbach 2014](#)).

2. Formal preliminaries

Let $\mathcal{L}$ denote the first-order language of arithmetic, and let $\mathcal{L}_T = \mathcal{L} \cup \{T\}$ be the extension obtained by adding a unary truth predicate $T(x)$.

Definition 1 (Arithmetical Coding). For any formula $\varphi \in \mathcal{L}_T$, let $\ulcorner \varphi \urcorner$ denote its canonical Gödel number under a fixed primitive recursive coding scheme.

The standard axiomatization over Peano Arithmetic (PA) requires compositional truth clauses:

$$\text{PA} \models T(\ulcorner \varphi \wedge \psi \urcorner) \iff T(\ulcorner \varphi \urcorner) \wedge T(\ulcorner \psi \urcorner).$$

Theorem 2 (Semantic Triviality). *If a theory $\Sigma \supset \text{PA}$ satisfies unrestricted capture and release:*

$$\Sigma \vdash \varphi \iff T(\ulcorner \varphi \urcorner),$$

then Σ is inconsistent under classical logic.

Proof. By the diagonal lemma, construct a sentence λ such that $\text{PA} \vdash \lambda \iff \neg T(\ulcorner \lambda \urcorner)$. Applying release yields $\Sigma \vdash \lambda \rightarrow \neg \lambda$, from which $\Sigma \vdash \perp$ follows by reductio. $\square$

3. Applications

To demonstrate further analytical and empirical extensions, we examine the following components:

- The asymptotic formulation of the natural exponential base e via infinitesimal change of variables.
- Dynamic modeling using a Vector Autoregressive (VAR) framework.
- Comparative analysis of bivariate distributions under varying correlation regimes.

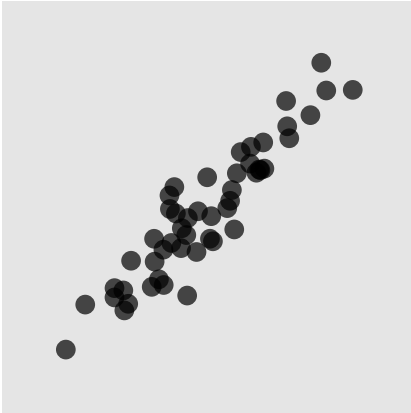
The natural exponential base e can be defined asymptotically. Setting $k = \frac{1}{n}$, as $n \rightarrow \infty$, we have $k \rightarrow 0$:

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = \lim_{k \rightarrow 0} (1 + k)^{\frac{1}{k}}, \quad \text{where } k = \frac{1}{n}$$

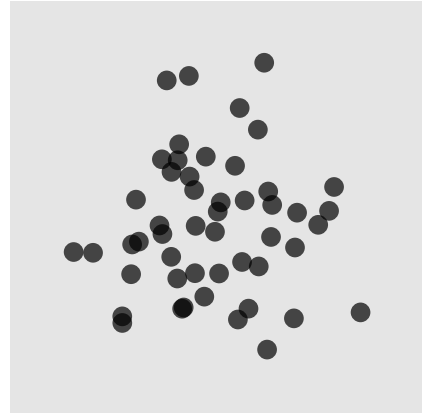
Furthermore, a Vector Autoregressive (VAR) model of order p is specified as:

$$Y_t = c + \sum_{i=1}^p A_i Y_{t-i} + \varepsilon_t$$

Finally, following the bivariate normal comparison in [Figure 1](#), [Figure 1a](#) illustrates the case where $\rho = 1$ while [Figure 1b](#) presents the uncorrelated case where $\rho = 0$:



(a) Bivariate Normal with $\rho = 1$



(b) Bivariate Normal with $\rho = 0$

Figure 1: Adding subfigures

References

- Halbach, Volker. 2014. *Axiomatic Theories of Truth*. Cambridge University Press. <https://doi.org/10.1017/cbo9781139696586>.
- Tarski, Alfred. 1936. "Der Wahrheitsbegriff in Den Formalisierten Sprachen." *Studia Philosophica* 1 : 261–405.